

Topic: Server Operating Systems and Their Features

Task 1: Comparison Table of Server OS

Criterion	Windows Server	Linux Server (Ubuntu/CentOS)	UNIX (FreeBSD/AIX)
Cost	High. You need to buy licenses for CPU cores and users.	Free. Open-source, you only pay if you want premium support.	Very high. Usually sold with expensive brand hardware.
Administration Difficulty	Medium. It has a familiar graphical interface (GUI) and tools.	Above medium. Mostly managed via text console (SSH).	High. Requires deep knowledge of command line and systems.
Application Area	Corporate networks, domain controllers, company emails.	Web servers, websites, cloud apps, database servers.	Banks, critical systems, mainframes, huge databases.
Advantages (Pros)	Easy to deploy, great support, good for office users.	Fast, flexible, secure, takes less system resources.	Maximum stability, legendary uptime, top-tier security.

Task 2: Situational Tasks (OS Selection)

Scenario A: Computer class in a school

Choice: Windows Server.

Justification: It is easy for a teacher to manage school computers from one place. We can quickly create accounts for students and block bad websites. Most school programs run on Windows perfectly.

Scenario B: Large online store

Choice: Linux Server (Ubuntu Server).

Justification: Linux is the best choice for web servers because it is very fast and stable under heavy traffic. It is free, which saves money for the business. Also, it is highly secure against hackers.

Scenario C: Banking system

Choice: UNIX.

Justification: Banks need 100% uptime and absolute security. UNIX systems are incredibly stable and can work for years without restarting. Banks have the budget for such expensive hardware and support.

Scenario D: Small office (10 people)

Choice: Windows Server Essentials.

Justification: A small office does not need a complex Linux setup. Windows Server is enough to share a printer, store shared office files, and manage 10 users easily.

Task 3: Analytical Question

Question: Why are requirements for stability and security higher for server OS than for desktop OS?

Answer: If a desktop computer crashes, only one user loses their unsaved work. But if a server crashes, hundreds of employees cannot work, and the company website goes down, causing financial loss. Servers hold critical and private data like passwords or credit cards, making them a main target for hackers. They also run 24/7 without breaks under constant network load. Therefore, a server operating system must be extremely stable and well-protected.